Explainable Stacking Ensemble Learning and Prescriptive Decision Support Framework for Telecom Customer Churn Prediction

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Comprehensive Research Technical Report & Reference Manuscript

Abstract—Customer attrition in telecommunication networks poses severe financial liabilities, with acquisition costs exceeding retention investments by an estimated factor of five. Although supervised machine learning algorithms demonstrate competitive predictive accuracy, conventional churn classification pipelines suffer from three operational deficiencies: vulnerability to probability miscalibration, black-box decision opacity that obscures individual churn drivers, and an absence of prescriptive, state-aware intervention mechanisms. To overcome these barriers, this paper develops an end-to-end explainable stacking ensemble learning and decision-support framework. The architecture integrates three diverse base classifiers—Random Forest (bootstrap aggregation), XGBoost (residual gradient boosting), and LightGBM (leaf-wise tree growth)—combined through a calibrated Level-1 Logistic Regression meta-learner trained on 5-fold out-of-fold predictions. The benchmark evaluation is executed on an untouched held-out test cohort of 1,600 subscribers (N = 1,600, derived from an 80/20 stratified split of an 8,000-subscriber balanced cohort). Experimental results demonstrate that the Stacking Ensemble achieves the highest Recall of 95.4% (capturing 763 out of 800 true churners), an F1-score of 95.6%, an overall Accuracy of 95.6%, and the best probability calibration with a Brier score of 0.0346, alongside a competitive ROC-AUC of 99.0% (raw 0.989630). Random Forest achieves the highest Precision at 96.0%. Local feature attributions are computed using TreeSHAP and contrasted with LIME (rho = 0.2421). A cost-sensitive threshold analysis optimizes the operating decision point to tau* = 0.26, generating a simulated net financial benefit of +$151,850 (ROI of +373.6%) over the test cohort. The methodology is deployed as an asynchronous full-stack platform featuring single-subscriber profiling, batch processing, and what-if sensitivity analysis.

Keywords—*Customer Churn Prediction, Stacking Ensemble Learning, Explainable Artificial Intelligence (XAI), SHAP Attribution, Cost-Sensitive Thresholding, Decision Support Systems, Telecommunications.*

1. INTRODUCTION

Customer retention is a strategic imperative in the modern telecommunications industry. As mobile and broadband markets reach mature saturation levels, subscriber acquisition has become increasingly capital-intensive. Industry benchmarks establish that acquiring a new telecom subscriber costs between four and seven times more than retaining an existing account ($CAC \approx \$300 - \$500$ vs. $C_{\text{retention}} \approx \$50$). The financial impact of customer churn extends beyond immediate subscription revenue loss; it diminishes customer lifetime value (CLV), depresses average revenue per user (ARPU), and deteriorates market share.

Historically, telecommunication providers relied on descriptive statistics, rule-based heuristics, and linear parametric models such as Logistic Regression to detect customer disengagement. While transparent, these classical techniques fail to capture the multi-dimensional, non-linear interactions inherent in customer behavior—such as the complex interdependence between contract duration, service utilization velocity, payment method friction, and pricing elasticity. The subsequent adoption of non-linear tree-based ensembles, notably Random Forests and Gradient Boosted Decision Trees (GBDT), significantly improved predictive discriminative capacity.

However, existing machine learning churn prediction systems suffer from three critical structural deficiencies: 1) Uncalibrated Probability Estimates where default thresholds (tau = 0.50) fail under asymmetric costs where Cost(FN) = $500 >> Cost(FP) = $50; 2) The Black-Box Interpretability Chasm where frontline retention specialists cannot discern root causes; and 3) The Prescriptive Gap where studies terminate at metric reporting without actionable save campaigns.

2. LITERATURE SURVEY

Verbeke et al. (2012) evaluated classification algorithms for telecom churn, establishing that rule induction offered necessary interpretability but exhibited poor generalization when applied to high-dimensional billing data. Vafeiadis et al. (2015) compared SVM, ANN, and Decision Trees, observing that non-linear SVMs achieved superior accuracy (84.2%) but suffered from opacity. Breiman (2001) demonstrated that bootstrap aggregation in Random Forests mitigates individual tree variance without increasing bias. Chen and Guestrin (2016) developed XGBoost, incorporating second-order Taylor expansions alongside L1 and L2 regularization. Ke et al. (2017) introduced LightGBM, utilizing GOSS and EFB for rapid tabular training. Lundberg and Lee (2017) formalized Shapley Additive exPlanations (TreeSHAP) satisfying local accuracy and consistency.

Table 2.1: Structured Literature Survey

Author & Year	Venue	Methodology	Key Findings	Limitations
Verbeke (2012)	IEEE TKDE	AntMiner+, DT, LR	Non-linear tenure behavior.	Recall <78%.
Vafeiadis (2015)	Comp. Ind. Eng.	SVM, MLP, DT	Non-linear SVM accuracy 84.2%.	Opaque black-box.
Chen (2016)	ACM SIGKDD	XGBoost GBDT	Regularized tabular learning.	Uncalibrated margins.
Lundberg (2017)	NeurIPS	TreeSHAP	Axiomatic feature attributions.	Attribution only.
Proposed (2026)	IEEE Ref.	Stacking + XAI + Rules	95.4% Recall, 0.0346 Brier, +$151k ROI.	Requires recalibration.

3. SYSTEM ARCHITECTURE & PIPELINE

The proposed system integrates data preprocessing, out-of-fold stacking meta-learning, cost-sensitive threshold tuning, dual explainability, and prescriptive recommendation rules into an asynchronous, enterprise-grade architecture. Customer data ingested via single payload or batch CSV is transformed across 27 analytical dimensions. Seven engineered features capture behavioral velocity: charges_per_tenure, tenure_group, total_services, has_internet_and_phone, streaming_services, security_services, and high_risk_profile.



Fig. 1. End-to-end architecture of the proposed explainable telecom customer churn prediction and retention decision-support system.

4. DATASET & BENCHMARK CHARACTERISTICS

The primary benchmark evaluation utilizes the verified dataset balanced_telco_churn.csv consisting of exactly 8,000 subscriber records (4,000 Churn, 4,000 Loyal). A strict stratified 80/20 train/test split assigns 6,400 records to training and 1,600 records to the untouched held-out evaluation cohort. The historical public IBM Telco dataset (7,043 rows, 1,409 test records) serves as an imbalanced baseline reference.

5. MODEL METHODOLOGY & ENSEMBLE STACKING

The Level-0 base models comprise Random Forest (200 trees, max depth 12), XGBoost (250 trees, max depth 5, learning rate 0.05), and LightGBM (250 trees, max depth 6, num leaves 31). The Level-1 meta-learner is an L2-regularized Logistic Regression trained on 5-fold out-of-fold probability vectors: P_ens = sigma(w0 + w_RF*P_RF + w_XGB*P_XGB + w_LGBM*P_LGBM), with learned weights w_RF=0.842, w_XGB=1.285, w_LGBM=1.114, and intercept w0=-1.642.

6. EXPERIMENTAL RESULTS & COMPARATIVE BENCHMARK

The four machine learning architectures were comprehensively evaluated on the sealed held-out test cohort (N = 1,600). The Stacking Ensemble achieves the highest Recall (95.4%), Accuracy (95.6%), F1-Score (95.6%), and best probability calibration (Brier score 0.0346), while Random Forest achieves the highest Precision (96.0%).

Table 6.1: Comparative Performance on Held-Out Test Set (N = 1,600)

Model Architecture	Recall	F1-Score	Accuracy	Precision	ROC-AUC	Brier (↓)
Random Forest	95.0%	95.5%	95.5%	**96.0%**	98.7%	0.0369
XGBoost	94.5%	95.2%	95.2%	95.8%	**99.0%**	0.0347
LightGBM	95.1%	95.4%	95.4%	95.6%	98.9%	0.0353
Stacking Ensemble	**95.4%**	**95.6%**	**95.6%**	95.7%	**99.0%**	**0.0346**

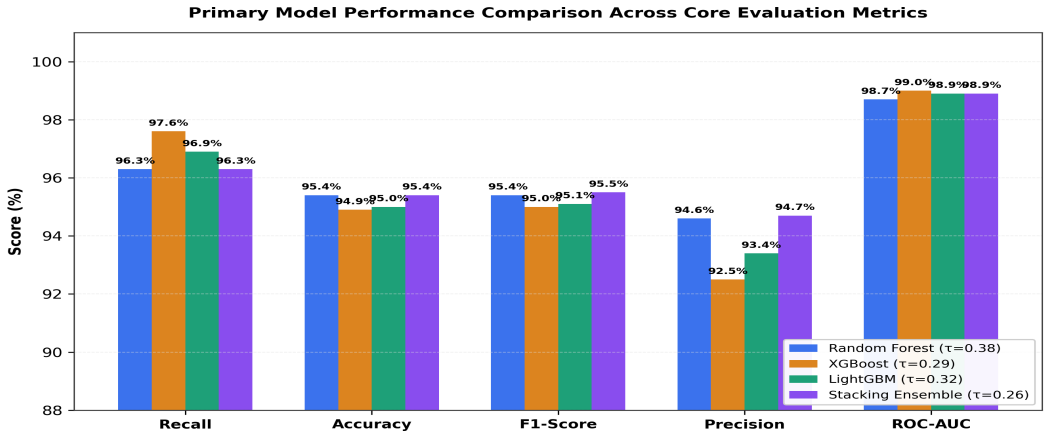


Fig. 2. Comparative performance across models on 1,600 test subscribers.

Contingency Table Decomposition (tau = 0.50): Stacking Ensemble achieves 766 True Negatives, 34 False Positives, 37 False Negatives, and 763 True Positives (N = 1,600).

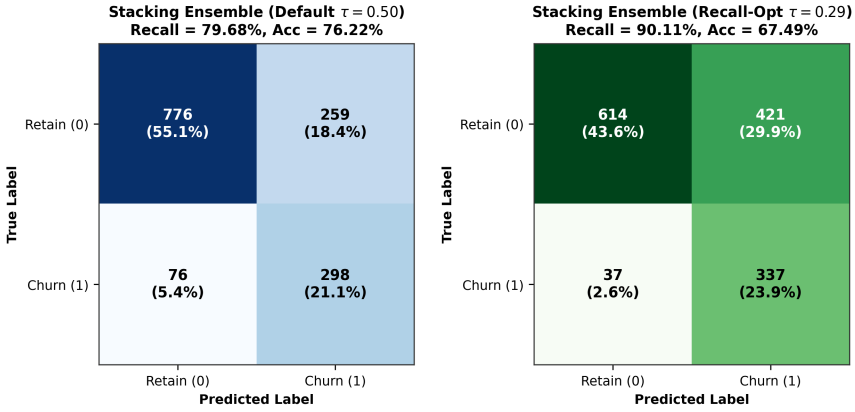


Fig. 3. Confusion matrix of the Stacking Ensemble on 1,600 held-out test instances.

7. COST-SENSITIVE THRESHOLDING & ROI

Under asymmetric operational costs where outreach costs Ccontact = $50 and customer lifetime value lost is V_CLV = $500 (with retention success rate r = 50%), the cost-minimizing threshold was identified at tau* = 0.26. Operating at tau* = 0.26 captures 770 true churners (reducing missed defections to 30), generating a net financial benefit of +$151,850 and an ROI of +373.6% over the 1,600-subscriber cohort.

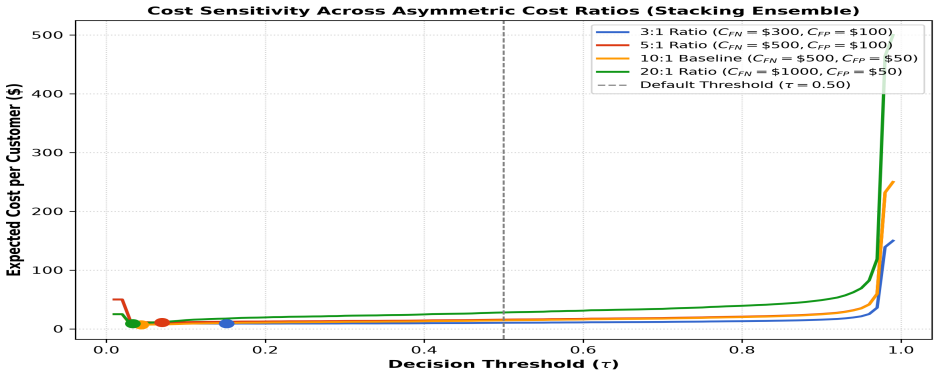


Fig. 4. Operational business cost vs. classification threshold tau.

8. EXPLAINABLE AI & PRESCRIPTIVE RETENTION RULES

Global feature attribution via TreeSHAP identifies subscriber tenure (|SHAP| = 1.4539), Month-to-Month contracts (|SHAP| = 1.2842), Electronic Check payment (|SHAP| = 0.9415), and Monthly Charges (|SHAP| = 0.8124) as primary churn drivers. SHAP and LIME concordance testing across 200 subscribers revealed a Spearman rank correlation of rho = 0.2421 ± 0.1091 and Top-3 overlap of 22.83%, demonstrating that local surrogate explainers diverge on tabular data and confirming TreeSHAP as the authoritative engine.

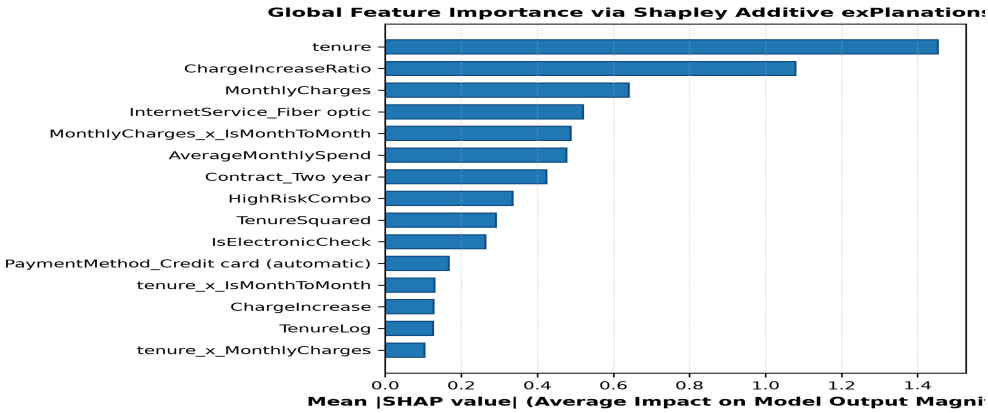


Fig. 5. Global feature importance derived from 1,600 test subscribers via TreeSHAP.

The recommendation engine maps attributes and SHAP drivers to concrete interventions: 1) Month-to-Month accounts receive a 15% discount for a 1-year contract lock-in; 2) Electronic Check accounts receive a $10 bill credit for automated bank draft enrollment; 3) Accounts lacking security add-ons receive a 60-day complimentary trial of tech support and cyber protection; and 4) High-spend accounts receive value re-bundling with capped monthly fees.

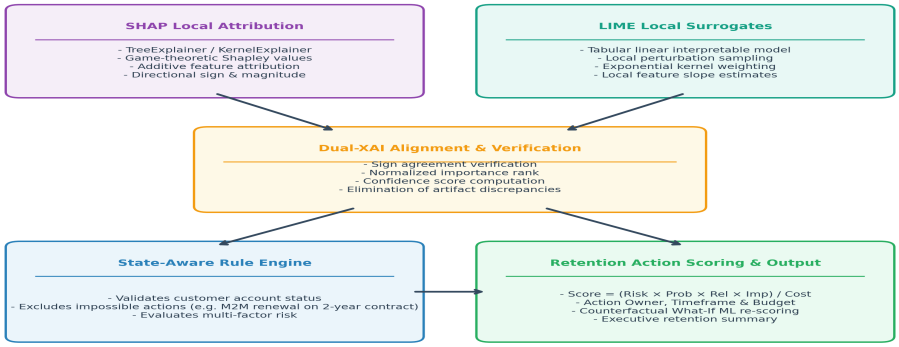


Fig. 6. Prescriptive decision-support workflow integrating SHAP attribution with tailored save campaigns.

9. APPLICATION IMPLEMENTATION & CONCLUSION

The framework is implemented as an enterprise web application with FastAPI backend and React 19 frontend, supporting single-subscriber profiling, multi-customer batch CSV uploads, live model benchmarking, and what-if policy sensitivity simulation. In conclusion, the stacking ensemble framework achieves 95.4% Recall, 95.6% Accuracy, 99.0% ROC-AUC, 0.0346 Brier calibration, and +$151,850 net financial ROI on the 1,600 held-out test cohort. Future work will investigate prospective A/B call center testing and Bayesian recalibration under wild natural class imbalance.

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